Tianxin Wei | Ph.D., University of Illinois Urbana-Champaign

Champaign, IL, USA

+1 217-954-3638 • twei10@illinois.edu • weitianxin.github.io

Google Scholar: scholar.google.com (2k+ citations)

Research Interests

I am an AI Research Scientist at Meta. Before that, I'm a Ph.D. student at the University of Illinois Urbana-Champaign. My research focuses on autonomous and self-improving LLM agents, with an emphasis on recursive self-improvement, reinforcement learning-based post-training, agent continual learning, and personalized agents. I am particularly interested in learning from continuous interactions.

Education

University of Illinois Urbana-Champaign — **Champaign, USA**

➢ *Ph.D. in Information Sciences* — *2021 Aug.–2026 Dec.(exp.)*

Advisor: Prof. Jingrui He, working with Prof. Hanghang Tong

University of Science and Technology of China — **Hefei, China**

➢ *B.S. in Computer Science* — *2016–2020*

School of the Gifted Young, Artificial Intelligence Talent Class

Work Experience

Meta AI — **Menlo Park, CA**

AI Research Scientist — *Aug 2026*

➢ Focus: Recursive self-improvement, agent post-training, continual learning, agent harness optimization, and personal agents.

Google DeepMind — **Mountain View, CA**

Student Researcher — *May-Aug 2025*

➢ Project: Evo-Memory: Benchmarking and Improving LLM Agent Test-time Learning with Self-Evolving Memory (In submission).
Advised by Wang-Cheng Kang, Noveen Sachdeva, Benjamin Coleman, Zhankui He, Chi Wang, Ed Chi, Fernando Pereira, and Derek Cheng

Meta AI — **Remote**

Research Intern — *Jan-May 2025*

➢ Project: CoFiRec: Coarse-to-Fine Tokenization for Generative Recommendation (RecSys'26).
Advised by Zhigang Hua, Shuang Yang

Amazon — **Palo Alto, CA**

Research Intern — *Jan.–May 2026*

➢ Project: Self-improving agents; context-to-harness skill compilation and sustained auto-harness evolution (Evo-Harness, Adaptive Auto-Harness, Harness Updating).

Amazon Rufus — **Palo Alto, CA**

Research Intern — *May-Aug 2024*

➢ Project: DiffKGW: Stealthy and Robust Diffusion Model Watermarking (TMLR).
Advised by Xianfeng Tang, Hui Liu, Chen Luo, and Qi He

Amazon Rufus — **Palo Alto, CA**

Research Intern — *May-Aug 2023*

➢ Project: Generative multimodal personalization; LLM for IR. (ICLR'24, WWW'24, ICML'24)
Advised by Xianfeng Tang, and Chen Luo

UT Austin — **Remote**

Research Intern — *Jan-May 2020*

➢ Project: Self-supervised graph learning in fintech, and recommender system (NeurIPS'22).
Advised by Prof. Yuning You, Tianlong Chen, and Zhangyang Wang

UCLA **Los Angeles, CA**
Visiting Scholar *May-Aug 2019*
- Project: Graph-based recommender systems; meta-path discovery (ICDM'20).
  Advised by Prof. Yizhou Sun, and Prof. Wei Wang

USTC **Hefei, China**
Research Intern *2019–2021*
- Project: Robustness and bias of recommender systems (KDD'21, SIGIR'21).
  Advised by Prof. Xiangnan He

Papers (* denotes equal contribution and † denotes supervision)

For the full list of publications, please refer to my Google Scholar profile.

Ongoing

2026 & 2025

- **Evo-Memory: Benchmarking LLM Agent Test-time Learning with Self-Evolving Memory**
  Tianxin Wei, Noveen Sachdeva, Benjamin Coleman, Zhankui He, Yuanchen Bei, Xuying Ning, Mengting Ai, Yunzhe Li, Jingrui He, Ed H. Chi, Chi Wang, Shuo Chen, Fernando Pereira, Wang-Cheng Kang, Derek Zhiyuan Cheng
  Featured in MIT Technology Review
- **Agentic Reasoning for Large Language Models: A Survey**
  Tianxin Wei, etc.
  Weekly #1 Huggingface Daily Papers
- **Code as Agent Harness**
  Xuying Ning*, Katherine Tieu*, Dongqi Fu*, **Tianxin Wei***, Zihao Li*, Yuanchen Bei*, etc.
  Daily #1 Huggingface Daily Papers
- **EvoHarness-RL: Learning Self-Evolving Runtime Harness for Long-Horizon LLM Agents**
- **Adaptive Auto-Harness: Sustained Self-Improvement for Agentic System Deployment on Open-Ended Task Streams**
  Zewen Liu, Zhan Shi, Yisi Sang, Bing He, Minhua Lin, **Tianxin Wei**, Dakuo Wang, Benoit Dumoulin, Wei Jin, Hanqing Lu
- **Heterogeneous Scientific Foundation Model Collaboration**
  Zihao Li, Jiaru Zou, Feihao Fang, Xuying Ning, Mengting Ai, **Tianxin Wei**, Sirui Chen, Xiyuan Yang, Jingrui He
  Daily #1 Huggingface Daily Papers
- **RRCM: Ranking-Driven Retrieval over Collaborative and Meta Memories for LLM Recommendation**
  Shijun Li, Wooseong Yang, Yu Wang, **Tianxin Wei**, Joydeep Ghosh

etc.

Conference Papers

- EMNLP 2026 Main

Evo-Harness: Context-to-Harness Skill Compilation for Self-Evolving Agents
Tianxin Wei, Zhan Shi, Minhua Lin, Bing He, Zewen Liu, Yisi Sang, Yuanchen Bei, Xuying Ning, Jiaru Zou, Ting-Wei Li, Xiao Lin, Yanjun Zhao, Chi Wang, Benoit Dumoulin, Dakuo Wang, Jingrui He, Hanqing Lu

- EMNLP 2026 Main

Harness Updating Is Not Harness Benefit: Disentangling Evolution Capabilities in Self-Evolving LLM Agents
Minhua Lin, Juncheng Wu, Zijun Wang, Zhan Shi, Yisi Sang, Bing He, Zewen Liu, **Tianxin Wei**, Zongyu Wu, Zhiwei Zhang, Dakuo Wang, Xiang Zhang, Benoit Dumoulin, Cihang Xie, Yuyin Zhou, Suhang Wang, Hanqing Lu

- EMNLP 2026 Main

ReContext: Recursive Evidence Replay as LLM Harness for Long-Context Reasoning
Yanjun Zhao*, Ruizhong Qiu*, **Tianxin Wei***, Yuanchen Bei, Zhining Liu, Lingjie Chen, Ismini Lourentzou, Hanghang Tong, Jingrui He

- EMNLP 2026 Main

ALERT: Zero-shot LLM Jailbreak Detection via Internal Discrepancy Amplification
Xiao Lin, Philip Li, Zhichen Zeng, Ting-Wei Li, **Tianxin Wei**, Xuying Ning, Gaotang Li, Yuzhong Chen, Hanghang Tong

- EMNLP 2026 Findings

CoReflect: A Reflective Co-Evolution Framework for Improving Conversational Evaluation
Yunzhe Li, Richie Yueqi Feng, **Tianxin Wei**, Chin-Chia Hsu

➢ EMNLP 2026 Main

Geometric-disentanglement Language Model Unlearning

Duo Zhou, Yuji Zhang, **Tianxin Wei**, Ruizhong Qiu, Ke Yang, Xiao Lin, Cheng Qian, Jingrui He, Hanghang Tong, ChengXiang Zhai, Heng Ji, Huan Zhang

➢ EMNLP 2026 Main

Subspace Alignment for Vision-Language Model Test-time Adaptation

Zhichen Zeng, Wenxuan Bao, Xiao Lin, Ruizhong Qiu, **Tianxin Wei**, Xuying Ning, Yuchen Yan, Chen Luo, Monica Xiao Cheng, Jingrui He, Hanghang Tong

➢ COLM 2026

Inference Scaling of LLM Ensembling: Bridging Token Spaces with Token Translation

Tianxin Wei, Zhichen Zeng, Ruizhong Qiu, Zhining Liu, Xuying Ning, Xinrui He, Wenxuan Bao, Xiao Lin, Qi He, Xianfeng Tang, Hanghang Tong, Jingrui He

➢ RecSys 2026

CoFiRec: Coarse-to-Fine Tokenization for Generative Recommendation

Tianxin Wei[*], Xuying Ning[*], Xuxing Chen, Ruizhong Qiu, Yupeng Hou, Yan Xie, Shuang Yang, Zhigang Hua, Jingrui He

➢ COLM 2026

Beyond Heuristics: A New Paradigm for Large Language Models Structured Pruning

Mengting Ai, **Tianxin Wei**, Sirui Chen, Jingrui He

➢ COLM 2026

EvoSelect: Data-Efficient LLM Evolution for Targeted Task Adaptation

Ting-Wei Li, Sirui Chen, Jiaru Zou, Yingbing Huang, **Tianxin Wei**, Jingrui He, Hanghang Tong

➢ COLM 2026

Reinforcement Routing for Mixtures of LoRAs in Parameter-Efficient LLM Finetuning

Ruizhong Qiu, Hanqing Zeng, Yinglong Xia, Yiwen Meng, Ren Chen, Jiarui Feng, Dongqi Fu, Qifan Wang, Jiayi Liu, Jun Xiao, Xiangjun Fan, Benyu Zhang, Hong Li, Zhining Liu, Hyunsik Yoo, Zhichen Zeng, **Tianxin Wei**, Hanghang Tong

➢ ICLR 2026

MC-Search: Evaluating and Enhancing Multimodal Agentic Search with Structured Long Reasoning Chains

Xuying Ning[*], Dongqi Fu[*], **Tianxin Wei**[*†], Mengting Ai, Jiaru Zou, Ting-Wei Li, Hanghang Tong, Yada Zhu, Hendrik Hamann, Jingrui He

Oral (Top 1%)

➢ ACL 2026 Main

AdaFuse: Adaptive Ensemble Decoding with Test-Time Scaling for LLMs

Chengming Cui[*], **Tianxin Wei**[*†], Ziyi Chen, Ruizhong Qiu, Zhichen Zeng, Zhining Liu, Xuying Ning, Duo Zhou, Jingrui He

➢ ACL 2026 Findings

PAPERMIND: Benchmarking Agentic Reasoning and Critique over Scientific Papers in Multimodal LLMs

Yanjun Zhao[*], **Tianxin Wei**[*†], Jiaru Zou, Xuying Ning, Yuanchen Bei, Lingjie Chen, Simmi Rana, Wendy H. Yang, Hanghang Tong, Jingrui He

➢ ACL 2026 Main

Mem-Gallery: Benchmarking Multimodal Long-Term Conversational Memory for MLLM Agents

Yuanchen Bei, **Tianxin Wei**, Xuying Ning, Yanjun Zhao, Zhining Liu, Xiao Lin, Yada Zhu, Hendrik Hamann, Jingrui He, Hanghang Tong

➢ ICLR 2026

Seeing but Not Believing: Probing the Disconnect Between Visual Attention and Answer Correctness in VLMs

Zhining Liu, Ziyi Chen, Hui Liu, Chen Luo, Xianfeng Tang, Suhang Wang, Jingying Zeng, Zhenwei Dai, Zhan Shi, **Tianxin Wei**, Hanqing Lu, Benoit Dumoulin, Hanghang Tong

➢ AAAI 2026

Panda: Test-Time Adaptation with Negative Data Augmentation

Ruxi Deng, Wenxuan Bao, Tianxin Wei, Jingrui He

➢ KDD 2025

Connecting Domains and Contrasting Samples: A Ladder for Domain Generalization

Tianxin Wei*, Yifan Chen*, Xinrui He, Wenxuan Bao, Jingrui He

➢ KDD 2025

ResMoE: Residual Approximation for Compressing Mixture-of-Experts in Large Language Models

Mengting Ai*, **Tianxin Wei***†, Yifan Chen*, Zhichen Zeng, Ritchie Zhao, Girish Varatkar, Bita Darvish Rouhani, Xianfeng Tang, Hanghang Tong, Jingrui He

➢ NeurIPS 2025

CLIMB: Class-imbalanced Learning Benchmark on Tabular Data

Zhining Liu, Zihao Li, Ze Yang, **Tianxin Wei**, Jian Kang, Yada Zhu, Hendrik Hamann, Jingrui He, Hanghang Tong

➢ ICCV 2025

Latte: Collaborative Test-Time Adaptation of Vision-Language Models in Federated Learning

Wenxuan Bao, Ruxi Deng, Ruizhong Qiu, **Tianxin Wei**, Hanghang Tong, Jingrui He

➢ EMNLP 2025 (Findings)

Learning to Instruct: Fine-Tuning a Task-Aware Instruction Optimizer for Black-Box LLMs

Yunzhe Qi, Jinjin Tian, Tianci Liu, Ruirui Li, **Tianxin Wei**, Hui Liu, Xianfeng Tang, Monica Xiao Cheng, Jingrui He

➢ CIKM 2025

PyG-SSL: A Graph Self-Supervised Learning Toolkit

Lecheng Zheng, Baoyu Jing, Zihao Li, Zhichen Zeng, **Tianxin Wei**, Mengting Ai, Xinrui He, Lihui Liu, Dongqi Fu, Jiaxuan You, Hanghang Tong, Jingrui He

➢ ACL 2025 (Main)

SelfElicit: Your Language Model Secretly Knows Where is the Relevant Evidence

Zhining Liu, Rana Ali Amjad, Ravinarayana Adkathim, **Tianxin Wei**, Hanghang Tong

➢ ACL 2025 (Findings)

LLM-Forest: Ensemble Learning of LLMs with Graph-Augmented Prompts for Data Imputation

Xinrui He, Yikun Ban, Jiaru Zou, **Tianxin Wei**, Curtiss Cook, Jingrui He

➢ UAI 2025

i^{2}VAE: Interest Information Augmentation with Variational Regularizers for Cross-Domain Sequential Recommendation

Xuying Ning, Wujiang Xu, **Tianxin Wei**, Xiaolei Liu

➢ ICML 2025

Graph4MM: Weaving Multimodal Learning with Structural Information

Xinrui He, Yikun Ban, Dongqi Fu, **Tianxin Wei**, Wujiang Xu, Jingrui He

➢ ICML 2025

Breaking Silos: Adaptive Model Fusion Unlocks Better Time Series Forecasting

Zhining Liu, Ze Yang, Xiao Lin, Ruizhong Qiu, **Tianxin Wei**, Yada Zhu, Hendrik Hamann, Jingrui He, Hanghang Tong

➢ ICLR 2024

Towards Unified Multi-Modal Personalization: Large Vision-Language Models for Generative Recommendation and Beyond

Tianxin Wei, Bowen Jin, Ruirui Li, Hansi Zeng, Zhengyang Wang, Jianhui Sun, Qingyu Yin, Hanqing Lu, Suhang Wang, Jingrui He, Xianfeng Tang

➢ KDD 2024

Neural Collaborative Filtering Bandits via Meta Learning

Yikun Ban, Yunzhe Qi, **Tianxin Wei**, Lihui Liu, Jingrui He

➢ AAAI 2024

TAU: Trajectory Data Augmentation with Uncertainty for Next POI Recommendation

Zhuang Zhuang, **Tianxin Wei**, Lingbo Liu, Heng Qi, Yanming Shen, Baocai Yin

➢ ICML 2024

Graph Mixup on Approximate Gromov–Wasserstein Geodesics

Zhichen Zeng, Ruizhong Qiu, Zhe Xu, Zhining Liu, Yuchen Yan, **Tianxin Wei**, Lei Ying, Jingrui He, Hanghang Tong

➢ ICML 2024

Language Models as Semantic Indexers

Bowen Jin, Hansi Zeng, Guoyin Wang, Xiusi Chen, **Tianxin Wei**, Ruirui Li, Zhengyang Wang, Zheng Li, Yang Li, Hanqing Lu, Suhang Wang, Jiawei Han, Xianfeng Tang

➢ WWW 2024

Scalable and Effective Generative Information Retrieval

Hansi Zeng, Chen Luo, Bowen Jin, Sheikh Muhammad Sarwar, **Tianxin Wei**, Hamed Zamani

➢ ICML 2023

NTK-approximating MLP Fusion for Efficient Language Model Fine-tuning

Tianxin Wei[*†], Zeming Guo[*], Yifan Chen[*], Jingrui He

➢ NeurIPS 2023

Adaptive Test-Time Personalization for Federated Learning

Wenxuan Bao[*], **Tianxin Wei**[*], Haohan Wang, Jingrui He

➢ NeurIPS 2023

Meta-Learning with Neural Bandit Scheduling

Yunzhe Qi, Yikun Ban, **Tianxin Wei**, Jiaru Zou, Huaxiu Yao, Jinrui He

➢ NeurIPS 2022

Augmentations in Hypergraph Contrastive Learning: Fabricated and Generative

Tianxin Wei[*], Yuning You[*], Tianlong Chen, Yang Shen, Jingrui He, Zhangyang Wang

➢ KDD 2022

Comprehensive Fair Meta-learned Recommender System

Tianxin Wei, Jingrui He

➢ KDD 2021

Model-Agnostic Counterfactual Reasoning for Eliminating Popularity Bias in Recommender System

Tianxin Wei, Fuli Feng, Jiawei Chen, Ziwei Wu, Jinfeng Yi, Xiangnan He

➢ SIGIR 2021

Causal Intervention for Leveraging Popularity Bias in Recommendation

Yang Zhang, Fuli Feng, Xiangnan He, **Tianxin Wei**, Chonggang Song, Guohui Ling, Yongdong Zhang

Best Paper Honorable Mention

➢ ICDM 2020

Fast Adaptation for Cold-start Collaborative Filtering with Meta-learning

Tianxin Wei, Ziwei Wu, Ruirui Li, Ziniu Hu, Fuli Feng, Xiangnan He, Yizhou Sun, Wei Wang

Journal Articles

➢ TMLR

DiffKGW: Stealthy and Robust Diffusion Model Watermarking

Tianxin Wei, Ruizhong Qiu, Yifan Chen, Yunzhe Qi, Jiacheng Lin, Wenju Xu, Sreyashi Nag, Ruirui Li, Hanqing Lu, Zhengyang Wang, Chen Luo, Hui Liu, Suhang Wang, Jingrui He, Qi He, Xianfeng Tang

➢ TMLR

A Survey of Agent Memory in the Second Half: Towards Self-Evolving and Long-Horizon Agents

Wei-Chieh Huang, Weizhi Zhang, Yueqing Liang, Yuanchen Bei, Yankai Chen, et al., **Tianxin Wei**, et al.

➢ TMLR

Simple is Better than Complex: A Representation-centric Perspective for Prompting-based Vision-Language Fusion

Yujia Yin, Jinhong Ni, Renjie Wu, Hongji Li, **Tianxin Wei**, Zhong Li, Yifan Chen

➢ TMLR

Pave Your Own Path: Graph Gradual Domain Adaptation on Fused Gromov–Wasserstein Geodesics

Zhichen Zeng, Ruizhong Qiu, Wenxuan Bao, **Tianxin Wei**, Xiao Lin, Yuchen Yan, Tarek F. Abdelzaher, Jiawei Han, Hanghang Tong

➢ npj Artificial Intelligence

Flow Matching Meets Biology and Life Science: A Survey

Zihao Li, Zhichen Zeng, Xiao Lin, Feihao Fang, Yanru Qu, Zhe Xu, Zhining Liu, Xuying Ning, **Tianxin Wei**, Ge Liu, Hanghang Tong, Jingrui He

➢ IEEE Data Engineering Bulletin

Graph Contrastive Learning: An Odyssey towards Generalizable, Scalable and Principled Representation Learning on Graphs

Yan Han, Yuning You, Wenqing Zheng, Scott Hoang, **Tianxin Wei**, Majdi Hassan, Tianlong Chen, Ying Ding, Yang Shen, Zhangyang Wang

Community Service

PC/Reviewer: ICML 2022–2026, NeurIPS 2022–2026, AAAI 2023–2026, ICLR 2023–2026, KDD 2023–2026, CIKM 2021–2024,

Journal Reviewer: Nature Communications, TPAMI, TOIS, TKDE, Information Fusion, TMLR, TKDD

Awards

2026: Google Student Award ($10k)
2026: Tinker Research Grant ($5k)
2025: KDD Outstanding Reviewer
2024: Amazon Fellowship ($40k)
2024: University Nomination (Top 3) for Apple Scholars in AI/ML
2023: NeurIPS Scholar Award
2022: NeurIPS Scholar Award
2023: ICML Grant Award
2023: UIUC Conference Presentation Award
2021: Best Paper Honorable Mention, SIGIR
2017–2020: USTC Outstanding Undergraduate Scholarship
2016: USTC Freshman Undergraduate Scholarship